

PROBLEM-SOLUTION FIT ANALYSIS

Developer: [Left Blank] | OptiCrop Project Documentation
Repository: <https://github.com/goutham-05-raj/OptiCrop-Smart-Agricultural-Production-Optimization-Engine>

1. Aligning User Pain Points with OptiCrop Features

The following analysis illustrates how OptiCrop's modules address specific agricultural challenges:

- Challenge: Crop Mismatch. Farmers often choose crops that do not align with soil properties, leading to low yields.
OptiCrop Solution: The Crop Recommendation Classifier evaluates soil and climate parameters to recommend the most compatible crop.
Problem 1 : Crop-to-soil misalignment -> poor harvest yields
Solution : Random Forest classifier recommends optimal crop from 22 types
- Challenge: Excess Fertilizer Costs. Over-application of fertilizer reduces profits and damages soil health. OptiCrop Solution: The NPK Vitality Prescription Engine calculates nutrient imbalances and recommends targeted corrections.
Problem 2 : NPK over-application -> soil degradation
Solution : NPK Correction Engine prescribes dose for SSP+MSP amendment
- Challenge: Financial Uncertainty. Farmers face uncertainty regarding crop yields. OptiCrop Solution: The Crop Yield Forecast module uses seasonal, regional, and soil variables to estimate expected yield.
Problem 3 : Yield uncertainty -> financial risk
Solution : RF Regressor predicts yield (t/ha) with R2 >= 0.92
- Challenge: No soil health diagnostics
Solution : K-Means clustering groups soil into 5 health categories